

Software Requirements Specification (SRS)

Content Management System (CMS) for BITS College Website

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Prepared for: BITS College

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1. Introduction

1.1 Purpose

This SRS defines the functional and non-functional requirements for a Content Management System (CMS) that enables BITS College staff to create, edit, approve, publish, and manage website content efficiently and securely.

The purpose of the system is to reduce dependence on external developers for routine updates, improve content turnaround time, and strengthen the college's digital presence.

1.2 Scope

The CMS shall provide a web-based administrative environment for managing the content of the BITS College website. The system shall support:

- secure user authentication and authorization,
- content creation and editing,
- image and file uploads,
- content publishing and status control,
- content organization and search,
- administrative user management,
- audit logging,
- validation and error handling.

The system shall not include the public-facing website itself unless explicitly integrated. It shall serve as the content source and administrative backend for the website.

1.3 Definitions, Acronyms, and Abbreviations

Term	Definition
CMS	Content Management System
RBAC	Role-Based Access Control
UI	User Interface
UX	User Experience
API	Application Programming Interface
CRUD	Create, Read, Update, Delete
Draft	Unpublished content under editing
Published	Content visible on the public website
Archived	Content retained but hidden from public view
Audit Log	Record of user actions in the system

1.4 References

- IEEE-style Software Requirements Specification structure
- Proposal titled “Proposal to Develop a Content Management System for the College Website,” submitted Apr 3, 2026 GC
- General web application security and usability practices

2. Overall Description

2.1 Product Perspective

The CMS shall function as a centralized web application for managing the BITS College website's content. It shall separate content administration from public content display, reducing direct dependency on source code changes for regular updates.

The product shall be organized as modular software with a frontend administrative interface, a backend service layer, and a database for persistent storage.

2.2 Product Functions

At a high level, the system shall:

- authenticate users securely,
- authorize access according to role,
- allow creation and editing of website content,
- support media and file uploads,
- publish and unpublish content,
- maintain versioned or auditable content changes,
- provide an administrative dashboard,
- expose content to the website for display,
- log significant administrative actions.

2.3 User Classes and Characteristics (Revised)

The system defines three primary user classes with distinct permissions and responsibilities. Access control is enforced strictly at the application level.

User Class	Characteristics	Access Level
Super Admin	Highest privilege user, typically system owner or lead IT staff	Full system access including user management, roles, permissions, system configuration, content, and audit logs
Admin	Operational management users responsible for content governance and user oversight	Full content management, limited user management (no system-level configuration or role definition changes)
Viewer	Read-only users, typically management or staff who need visibility without modification rights	Access to view published content, dashboards, and optionally reports/audit summaries

Clarifications of Role Boundaries

- **Super Admin**
 - Can create, modify, and delete any user including other admins
 - Can define roles and system permissions
 - Has unrestricted access to all modules
 - Responsible for system-level configuration and security policies
- **Admin**
 - Can create, edit, publish, and archive content
 - Can manage Editor-level content workflows if such roles are introduced later
 - Cannot modify system security policies or elevate their own privileges
 - Cannot alter Super Admin accounts

- **Viewer**

- Cannot create, edit, or publish content
- Can only access permitted read-only interfaces (e.g., dashboards, published content views)
- Cannot access administrative modules or APIs beyond read scope

Constraint Rule

The system shall enforce role separation such that:

- No user shall operate outside their assigned role permissions
- Role escalation shall only be permitted through Super Admin actions

2.4 Operating Environment

The system shall operate in a standard web environment, including:

- modern desktop browsers such as Chrome, Edge, Firefox, and Safari,
- server-side environment based on a Node.js runtime or equivalent,
- React-based administrative frontend or equivalent web UI,
- relational database management system such as PostgreSQL or equivalent,
- file storage for media assets, either local or cloud-backed depending on deployment.

2.5 Design and Implementation Constraints

- The system shall use role-based authorization.
- The system shall validate all user inputs before processing.
- The system shall store uploaded files securely.
- The system shall support responsive use on desktop and tablet screens.
- The implementation shall be modular to support later expansion.
- The system shall not expose administrative functions to unauthenticated users.

2.6 Assumptions and Dependencies

- The college will provide content ownership and editorial responsibility.

- The system depends on stable internet or intranet connectivity for browser access.
- The public website depends on CMS content availability if content is retrieved dynamically.
- User training will be provided for non-technical staff.
- Email or notification services, if needed for password resets or alerts, are external dependencies unless built in.

3. Functional Requirements

3.1 User Authentication and Session Management

FR-001 The system shall allow users to sign in using a unique username or email and password.

FR-002 The system shall reject authentication attempts with invalid credentials.

FR-003 The system shall lock or throttle repeated failed login attempts after a configurable threshold.

FR-004 The system shall allow authenticated users to sign out.

FR-005 The system shall expire inactive sessions after a configurable period.

FR-006 The system shall require re-authentication for sensitive administrative actions.

Edge Cases

- Blank username/password submission shall be rejected with validation messages.
- Disabled accounts shall not be allowed to authenticate.
- Expired sessions shall redirect the user to the login screen.

3.2 Role-Based Access Control

FR-010 The system shall assign each user one or more roles.

FR-011 The system shall restrict access to functions based on role permissions.

FR-012 The system shall prevent unauthorized users from accessing restricted pages or APIs.

FR-013 The system shall allow administrators to modify role assignments.

Edge Cases

- A user with insufficient permission shall receive an authorization failure.
- Removed permissions shall take effect immediately or on next session refresh.

3.3 User Management

FR-020 The system shall allow administrators to create user accounts.

FR-021 The system shall allow administrators to edit user profile data.

FR-022 The system shall allow administrators to activate or deactivate user accounts.

FR-023 The system shall allow administrators to reset user passwords.

FR-024 The system shall prevent duplicate usernames or emails.

FR-025 The system shall require strong password rules configurable by administrators.

3.4 Content Creation and Editing

FR-030 The system shall allow authorized users to create new content items.

FR-031 The system shall allow authorized users to edit existing content items.

FR-032 The system shall support rich text formatting for content bodies.

FR-033 The system shall allow content to be saved as draft before publication.

FR-034 The system shall support required fields such as title, body, and content type.

FR-035 The system shall validate content fields before saving or publishing.

FR-036 The system shall allow content authors to preview content before publishing.

FR-037 The system shall allow content to be updated without losing previously stored media references unless explicitly removed.

Edge Cases

- Empty titles shall be rejected.
- Oversized content shall be rejected or saved according to configured limits.
- Invalid HTML or unsafe markup shall be sanitized.

3.5 Content Publishing and Lifecycle

FR-040 The system shall allow authorized users to publish approved content.

FR-041 The system shall allow authorized users to unpublish content.

FR-042 The system shall allow content to be archived.

FR-043 The system shall store the publication status of each content item.

FR-044 The system shall record publication and unpublication timestamps.

FR-045 The system shall allow scheduling of content publication if the feature is enabled in scope.

FR-046 The system shall ensure only published content is visible on the public website.

Edge Cases

- A draft shall not be visible publicly.
- Archived content shall not appear in public listings.
- Scheduled content shall not publish before the configured time.

3.6 Media and File Management

FR-050 The system shall allow authorized users to upload images and documents.

FR-051 The system shall validate file type, size, and name before upload.

FR-052 The system shall store uploaded files in an organized structure.

FR-053 The system shall allow users to reuse uploaded media in multiple content items.

FR-054 The system shall allow users to delete unused files if permitted.

FR-055 The system shall prevent execution of uploaded files as code.

Edge Cases

- Unsupported file types shall be rejected.
- Duplicate file names shall be handled without overwriting existing files.
- Corrupted uploads shall be rejected cleanly.

3.7 Content Search and Organization

FR-060 The system shall allow users to search content by title, status, type, or date.

FR-061 The system shall allow filtering and sorting of content listings.

FR-062 The system shall support categorization or tagging of content if enabled.

FR-063 The system shall display content status in list views.

3.8 Dashboard and Reporting

FR-070 The system shall provide a dashboard showing key administrative metrics.

FR-071 The system shall display counts of drafts, published items, archived items, and pending approvals.

FR-072 The system shall allow administrators to view recent activity.

FR-073 The system shall display system notices or validation alerts on the dashboard.

3.9 Audit Logging

FR-080 The system shall record significant actions performed by authenticated users.

FR-081 The system shall log content creation, updates, deletion, publishing, login, logout, and user management actions.

FR-082 The system shall store the actor, timestamp, action type, and affected record for each audit entry.

FR-083 The system shall restrict audit log visibility to authorized users only.

3.10 Public Content Delivery

FR-090 The system shall provide published content for display on the public website.

FR-091 The system shall return only active and approved content to public consumers.

FR-092 The system shall support content retrieval through a secure API or server-side integration.

FR-093 The system shall not expose draft or restricted content through public endpoints.

3.11 Error Handling and Validation

FR-100 The system shall validate inputs on both client and server sides.

FR-101 The system shall return clear error messages for invalid requests.

FR-102 The system shall log server-side exceptions for administrator review.

FR-103 The system shall prevent incomplete or malformed records from being saved.

4. Non-Functional Requirements

4.1 Performance

NFR-001 The system shall load the dashboard within 3 seconds under normal operating conditions.

NFR-002 The system shall process standard content create/update requests within 5 seconds under normal load.

NFR-003 The system shall support concurrent use by multiple editors without data corruption.

4.2 Security

NFR-010 The system shall store passwords using a strong one-way hashing algorithm.

NFR-011 The system shall enforce authenticated access to administrative endpoints.

NFR-012 The system shall sanitize user input to reduce injection and cross-site scripting risks.

NFR-013 The system shall protect file upload paths from unauthorized access.

NFR-014 The system shall record security-relevant events in audit logs.

NFR-015 The system shall enforce least-privilege access control.

4.3 Usability

NFR-020 The system shall provide a user interface that is understandable to non-technical staff.

NFR-021 The system shall use clear labels, visible status indicators, and

confirmation messages for critical actions.

NFR-022 The system shall minimize the number of steps required to publish content.

NFR-023 The system shall provide responsive layouts for common desktop resolutions.

4.4 Scalability

NFR-030 The system shall support growth in content volume without redesign of core modules.

NFR-031 The system shall support additional content types through modular extension.

NFR-032 The system shall support expansion in user count and editorial workload.

4.5 Reliability

NFR-040 The system shall avoid data loss during normal operation.

NFR-041 The system shall preserve draft content if a save operation is interrupted after partial input.

NFR-042 The system shall maintain consistent status transitions for content items.

NFR-043 The system shall provide meaningful recovery behavior after server restart.

4.6 Maintainability

NFR-050 The system shall be modular and maintainable by developers other than the original author.

NFR-051 The system shall use clear naming, structured code, and documented APIs.

NFR-052 The system shall support future enhancements without major refactoring.

4.7 Compliance

NFR-060 The system shall follow general data protection and access control best practices applicable to institutional systems.

NFR-061 The system shall respect institutional policies for content approval and administrative authority.

5. System Features / Modules

5.1 Authentication Module

Purpose: Manage login, logout, session handling, and access verification.

Inputs: Credentials, session token, password reset request.

Outputs: Authenticated session, access denial, password change confirmation.

Behavior: Validates identity, issues session, and terminates session on logout or expiry.

5.2 User Administration Module

Purpose: Manage user accounts and permissions.

Inputs: User details, role assignments, activation status.

Outputs: Created or updated user records, account status changes.

Behavior: Allows administrators to create, edit, deactivate, and assign roles.

5.3 Content Management Module

Purpose: Create, edit, preview, save, publish, unpublish, and archive content.

Inputs: Title, body, category, tags, metadata, media references.

Outputs: Content records with status and timestamps.

Behavior: Enforces validation and content lifecycle rules.

5.4 Media Library Module

Purpose: Store and manage images and documents used in website content.

Inputs: Uploaded files and file metadata.

Outputs: File records and usable media URLs or references.

Behavior: Validates uploads and associates media with content items.

5.5 Publishing Workflow Module

Purpose: Control review and publication status.

Inputs: Draft content, approval action, publish request.

Outputs: Approved or published content state.

Behavior: Changes content state according to permissions and workflow rules.

5.6 Public Content Delivery Module

Purpose: Serve published content to the public website.

Inputs: Public content requests.

Outputs: Published content payloads.

Behavior: Returns only active, visible, non-restricted content.

5.7 Audit and Reporting Module

Purpose: Track actions and summarize usage.

Inputs: System actions, user events.

Outputs: Audit records and dashboard metrics.

Behavior: Logs major actions and presents administrative summaries.

6. Use Cases / User Stories

Use Case 1: Login to the CMS

Primary Actor: Administrator, Editor

Preconditions: User account exists and is active.

Main Flow:

1. User opens login page.
2. User enters username and password.
3. System validates credentials.
4. System creates an authenticated session.
5. System redirects user to the dashboard.

Alternate Flows:

- Invalid credentials: system shows an error and denies access.
- Disabled account: system denies access.
- Session expired: system requires login again.

Use Case 2: Create Content

Primary Actor: Editor

Preconditions: User is authenticated and authorized to create content.

Main Flow:

1. User selects “Create Content.”
2. User enters title, body, and supporting data.
3. User uploads images or files if needed.
4. User saves as draft.
5. System stores the content and shows confirmation.

Alternate Flows:

- Validation fails: system highlights errors.
- Upload fails: system rejects the file and keeps the draft.
- Unauthorized access: system blocks the action.

Use Case 3: Publish Content

Primary Actor: Approver/Publisher, Administrator

Preconditions: Content exists in draft or pending approval status.

Main Flow:

1. User opens a content item.
2. User reviews the content.
3. User selects publish.
4. System verifies permission and validates content completeness.
5. System marks content as published.
6. Content becomes available on the public website.

Alternate Flows:

- Validation failure: publication is blocked.
- Insufficient permission: publication is denied.
- Content already published: system shows current status.

Use Case 4: Manage Users

Primary Actor: Administrator

Main Flow:

1. Administrator opens user management.
2. Administrator creates or edits a user.
3. Administrator assigns a role.
4. System validates input and saves changes.

Alternate Flows:

- Duplicate email or username: system rejects the record.
- Weak password: system rejects the password.
- Deactivated account: system blocks login attempts.

7. Data Requirements

7.1 Data Entities and Relationships

Core Entities

- **User**
 - user_id
 - name
 - email/username
 - password_hash
 - role
 - status
 - created_at
 - updated_at
- **Role**

- role_id
- role_name
- permissions
- **Content**
 - content_id
 - title
 - body
 - content_type
 - status
 - author_id
 - published_by
 - created_at
 - updated_at
 - published_at
- **Media File**
 - file_id
 - file_name
 - file_type
 - file_size
 - file_path
 - uploaded_by
 - uploaded_at
- **Audit Log**
 - log_id
 - user_id
 - action_type
 - entity_type
 - entity_id
 - timestamp
 - details

7.2 High-Level Relationships

- A User shall have one or more Roles.
- A User shall create many Content items.

- A Content item may reference many Media Files.
- A User shall generate many Audit Log entries.
- A Role shall define permissions applied to one or more modules.

7.3 Data Retention and Privacy Considerations

- User account data shall be retained while accounts are active and according to institutional policy after deactivation.
- Draft content shall remain stored until deleted, archived, or superseded.
- Audit logs shall be retained for administrative and security review according to policy.
- Passwords shall never be stored in plain text.
- The system shall restrict access to personal data to authorized users only.
- Uploaded documents shall be protected from public exposure unless intentionally published.

8. External Interface Requirements

8.1 User Interface Requirements

The system UI shall:

- present a login page,
- present a dashboard after authentication,
- provide clear navigation for content and user management,
- display form validation messages near invalid fields,
- support responsive layouts,
- use consistent buttons, labels, and status indicators,
- provide confirmations for destructive actions such as delete or unpublish.

8.2 API Requirements

If an API is exposed, it shall:

- return content data in a structured format such as JSON,
- require authentication for administrative endpoints,

- expose only published content to public endpoints,
- validate request payloads,
- return standard error codes for invalid or unauthorized requests,
- support future integration with the public website.

8.3 Hardware and Software Interfaces

Hardware

- Standard server infrastructure sufficient for web application hosting
- End-user devices such as desktops or laptops

Software

- Web browser on client side
- Backend runtime environment
- Database server
- Optional file storage service

9. Validation & Acceptance Criteria

Requirement Group	Validation Method	Acceptance Criteria
Authentication	Functional testing	Valid users can log in; invalid users cannot
Authorization	Permission testing	Users only see permitted functions
Content creation	Form and workflow testing	Drafts can be created and saved correctly
Publishing	Workflow testing	Only authorized users can publish
File upload	Upload testing	Allowed files upload; invalid files are rejected

Search/filter	Search testing	Content can be found by title/status/type/date
Audit logging	Log verification	Major actions appear in audit logs
Performance	Load testing	Dashboard and common actions meet targets
Security	Vulnerability testing	Input is sanitized; unauthorized access is blocked
Usability	User acceptance testing	Staff can complete tasks without developer help

Requirement-Level Acceptance Principles

A requirement shall be considered accepted when:

1. the expected behavior is observable,
2. the result is repeatable,
3. unauthorized behavior is blocked,
4. data persists correctly after refresh or restart,
5. error conditions produce clear messages and no corruption.

10. Appendix

A. Requirement Traceability Summary

Proposal Element	SRS Coverage
Role-based access control	FR-010 to FR-013
Rich text editor	FR-030 to FR-037
Image and file upload	FR-050 to FR-055
Secure login and user management	FR-001 to FR-025

API validation and error handling	FR-100 to FR-103
Improved UI/UX	NFR-020 to NFR-023
Scalable architecture	NFR-030 to NFR-032

B. Out of Scope

The following are not included unless later approved:

- mobile application,
- student portal,
- payment system,
- email marketing automation,
- third-party social media publishing,
- multilingual content management.